

Hazem Hassan Aly

Generative AI Engineer | RAG, Agentic Systems & Backend Development

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PROFESSIONAL SUMMARY

Generative AI Engineer at the intersection of language, data, and intelligent systems. Focused on RAG, agentic workflows, NLP, and backend development, with an emphasis on grounded outputs, rigorous evaluation, and transparent system behavior.

EXPERIENCE & TRAINING

Backend Engineer & Data Science Lead | PM Accelerator - Fit My Day | Aug 2026 - Present (Expected Oct 2026)

- Leading backend architecture, database-schema design, and system-flow planning for an in-progress workout-planning and recommendation platform using FastAPI and Supabase.
- Defining data and recommendation requirements, coordinating technical decisions, and documenting implementation boundaries.

Agentic AI & Generative AI Trainee | Digital Egypt Pioneers Initiative (DEPI) | Jul 2026 - Present (Expected Dec 2026)

- Developing skills in LLMs, RAG, agentic and multi-agent systems, AI application development, deployment, and monitoring through structured labs and projects.

NLP Engineering Intern | Elevvo | Aug 2026 - Sep 2026

- Built four NLP projects spanning topic modeling, extractive QA, privacy-aware RAG talent search, and a guarded Text-to-SQL agent.
- Benchmarked QA on 100 SQuAD examples and improved RAG hit rate@3 from 0.500 to 1.000 on eight labeled queries.

Generative AI Intern | NeuralSeek | Jul 2026 - Aug 2026

- Independently built a Generative AI agent that analyzes business challenges and recommends practical solutions.

Data Analytics Trainee | National Telecommunication Institute (NTI) | Completed Sep 2025

AI Response Evaluator | Outlier | Nov 2024 - Jan 2025

- Evaluated English and occasional Arabic AI responses against rubrics for factual accuracy, instruction following, reasoning, and writing quality.

EDUCATION

B.Sc. in Computer & Communications Engineering | Oct 2022 - Expected July 2027

Alexandria University, Faculty of Engineering | Alexandria, Egypt

PROJECTS

Arabic Sentiment Classifier | *Python, AraBERT, TF-IDF, Hugging Face, Gradio*

- Built a three-class Arabic tweet classifier and improved held-out macro F1 from 0.802 to 0.860; published the model on Hugging Face and verified the Gradio interface locally.

RAG-Powered Talent Search | *LangChain, FAISS, BM25, Streamlit*

- Built a privacy-aware search workflow for 220 resumes using PII masking, local embeddings, hybrid retrieval, and evidence-grounded results.

Autonomous Text-to-SQL Agent | *Python, LangChain, Gemini, SQLGlot, SQLite, Streamlit*

- Built an agent that converts business questions into SQL, validates and executes read-only queries, self-corrects database errors, and displays an inspectable SQL audit trail.

TECHNICAL SKILLS

Generative AI & NLP: Python, LangChain, RAG, FAISS, BM25, Hugging Face Transformers, embeddings, prompt engineering, Gemini

Machine Learning & Evaluation: PyTorch, TensorFlow, scikit-learn, AI response evaluation, XGBoost, LightGBM, Pandas, NumPy

Backend & Data: FastAPI, Supabase, Node.js, REST APIs, SQL, MySQL, Docker

Applications: Streamlit, Gradio, React, Plotly

CERTIFICATES

AWS AI Practitioner Challenge | Udacity | Verified Course Completion | Jun 2026

Getting Started with Deep Learning | NVIDIA Deep Learning Institute | Certificate of Competency | Sep 2025